

Global spin phases quantize a ratio, not an absolute action scale

A classical spin sphere has globally consistent Hamiltonian dynamics for any positive symplectic area. Requiring a well-defined phase with a supplied action constant K imposes the integer condition $2s/K \in \mathbb{Z}$. It restricts the allowed sphere area relative to K , while leaving the common normalization free and allowing arbitrarily small local loop actions.

This is Q07's exploratory premise test, with explicit patch calculations. It uses established prequantization geometry; it claims neither novelty nor an accepted project result. Its purpose is to locate the physical premise behind a tempting global-consistency route to action selection.

1. A classical action defined in patches

Let the spin direction $\mathbf{n}$ lie on the unit sphere with polar coordinates (θ, ϕ) . Supply $s > 0$ with action units and define

$$\omega_s = s \sin \theta \, d\theta \wedge d\phi, \quad \int_{S^2} \omega_s = 4\pi s. \quad (1)$$

For any smooth real Hamiltonian H , define the vector field by $\iota_{X_H} \omega_s = -dH$. Nondegeneracy gives a unique smooth vector field; compactness gives its complete flow. Thus s may be any positive real value. There is no globally defined one-form with derivative ω_s , since the integral of an exact two-form on the closed sphere would vanish.

Instead use regular north and south potentials

$$\alpha_N = s(1 - \cos \theta) d\phi, \quad \alpha_S = -s(1 + \cos \theta) d\phi. \quad (2)$$

Each has derivative ω_s . On a simply connected part of the overlap,

$$\alpha_N - \alpha_S = 2s \, d\phi = d(2s\phi). \quad (3)$$

The local actions $S_U = \int [\alpha_U(\dot{n}) - H(n)] dt$ differ by an endpoint term. Their fixed-endpoint variations agree, yielding the same global Hamiltonian equation. The azimuth is not single-valued on the entire overlap; going around it changes $2s\phi$ by $4\pi s$. This constant ambiguity does not change the equations. Classical variational consistency requires the local derivatives to agree, not a globally single-valued real action value.

As a concrete clock, take $H = -\Omega s \cos \theta$, where Ω has inverse time units. The equation gives $\dot{\phi} = \Omega$, $\dot{\theta} = 0$. Multiplying s and H by the same positive factor leaves this physical-time motion intact.

2. The added phase requirement

Now supply $K > 0$ with action units and ask for phase transport compatible with the local factors $\exp(iS_U/K)$. On the overlap the transition phase, up to convention for its inverse, is

$$g_{NS}(\phi) = \exp(i2s\phi/K).$$

This must be single-valued when ϕ advances by 2π . Hence

$$\exp(i4\pi s/K) = 1 \quad \Longleftrightarrow \quad \frac{2s}{K} \in \mathbb{Z}. \quad (4)$$

The same condition follows by spanning a closed loop with a surface: two spanning surfaces differ by an integer number of spheres, so their geometric actions differ by $4\pi s$ times an integer. Equation (4) makes the exponent independent of this choice. For open paths the transport is a map between endpoint phase fibres, rather than a gauge-independent scalar.

The extra premise is a $U(1)$ phase with curvature proportional to ω_s/K . Equation (4) is its compatibility condition. The classical symplectic form alone supplies no such phase, no

interference rule and no K. Prequantization also leaves further state-space and measurement choices open.

3. Three decisive countertests

Common rescaling. If (s, H, K) satisfies the classical and phase conditions, so does $(\eta s, \eta H, \eta K)$ for every $\eta > 0$. Both the Hamiltonian flow and the dimensionless phase remain unchanged. Thus even phase consistency fixes a ratio rather than an absolute dimensional constant.

Local contraction. With $H=0$, the geometric action around a latitude bounding a north-pole cap is

$$A(\theta_0) = 2\pi s(1 - \cos \theta_0) \longrightarrow 0 \quad (\theta_0 \downarrow 0). \quad (5)$$

These are trial loops, not nonconstant solutions for $H=0$. They test local action resolution: quantization of the total sphere flux allows arbitrarily small nonzero cap actions. It does not impose a minimum polygonal cell or invalidate ordinary time subdivision. At fixed K and $s>0$, (4) instead restricts the allowed total sphere area, giving $s \geq K/2$ in that family.

Two coupled spheres. Take the product with form $\omega_{s_1} \oplus \omega_{s_2}$ and any smooth interaction, for example $H = Jn_1 \cdot n_2$ with J in energy units. It defines complete classical dynamics for arbitrary $s_1, s_2 > 0$, including $s_2/s_1 = \sqrt{2}$. A common prequantization phase constant would require

$$2s_1/K = m_1, \quad 2s_2/K = m_2, \quad m_1, m_2 \in \mathbb{Z}_{>0}. \quad (6)$$

Restriction to each sphere with the other held fixed proves these necessary conditions; product transition functions make them sufficient for this prequantization construction. Thus $s_2/s_1 = m_2/m_1$ must be rational. Classical coupling admits the irrational example, while a single-K phase bundle excludes it. The exclusion comes from the added phase structure, not from a failure of reciprocal Hamiltonian mechanics. Even a permitted rational ratio leaves infinitely many choices obtained by K replaced by K/n .

4. Consequence for the programme

Topology can turn a common phase premise into an arithmetic selection rule. That is a stronger compatibility requirement than Q06's reciprocal forces, and the irrational pair makes the difference explicit. It still leaves the absolute scale free and local actions continuous.

Park further patching examples. The next useful construction should test a physical interference record: can two classical wave paths coupled to a mechanical receiver establish a universal action-to-phase conversion, rather than a frequency-, amplitude- or apparatus-dependent calibration? Specify energy transfer and the measured record before using any quantum probability rule. This changes the question from another global bundle condition to the physical origin of the phase premise that (4) requires.

Source boundary

F. Bonechi, A. S. Cattaneo and M. Zabzine, *Geometric quantization and non-perturbative Poisson sigma model*, *Advances in Theoretical and Mathematical Physics* **10**, 683–712 (2006), arXiv:math/0507223v3, explicitly attributes its integrality condition to requirements on a path integral. Coverage here: one discovery query and its arXiv abstract; no full-proof or exact sphere-formula audit. The local potentials, variational patching and tests (4)–(6) are written out above. Independent review and a bounded librarian comparison are required before any ledger promotion. No numerical or symbolic verification scripts were used.